

AppleOrange

STAT380

2023-10-05

We consider logistic regression for the binary classification.

The Package `OpenImageR` and setting the directory

```
#### Horea Muresan, Mihai Oltean, Fruit recognition from images using deep learning,  
Acta Univ. Sapientiae, Informatica Vol. 10, Issue 1, pp. 26–42, 2018.  
#install.packages("OpenImageR")  
library(OpenImageR)
```

Set the Directory Location

```
dir_location<-"/Users/subhadippal/Desktop/STAT380_Fruit_Data_Small/"
```

All Functions

Functions Used in the Exercise

Extracting Color Features from a single Image

```
Extract_col_feature<-function(img){  
  Clr1<-summary(c(img[, , 1]))[-c(1,6)]  
  Clr2<-summary(c(img[, , 2]))[-c(1,6)]  
  Clr3<-summary(c(img[, , 3]))[-c(1,6)]  
  val<-c(Clr1,Clr2, Clr3)  
  #browser()  
  names(val)=paste0("Feature",1:length(val) )  
  return(val)  
}
```

Extracting Color Features from all the Images of a folder

```
ExtractFeature<-function(FolderName,                                img_path_dir="/Users/subhadippa
l/Downloads/Fruit-Images-Dataset-master/Training/",
                        details=TRUE){
  #browser()
  #img_path1="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Training/Orang
e/0_100.jpg"
  files_names<-list.files(paste0(img_path_dir,FolderName))
  val_color_feature=NULL

  for(FileIndex in 1:length(files_names)){
    img_path<-paste0(img_path_dir,FolderName,"/" , files_names[FileIndex])
    #print(img_path)
    img <- tryCatch(expr = {readImage(img_path)}),
                    error = function(e){
                      # (Optional)
                      # Do this if an error is caught...
                      return(NULL)
                    }
  )

  #browser()
  if(!is.null(img)){
    val_color_feature<-rbind(val_color_feature, c(Extract_col_feature(img)))
  }
  if(details){
    print(paste0("ImageData='",img_path, "' Loaded Succesfully. Feature Extraction comp
lete"))}
  }
  Y=rep(x = FolderName, dim(val_color_feature)[1])
  #val_color_feature
  return(data.frame(Y,val_color_feature))
}
```

Function to Create the Training Dataset

```
Create_Training_Data<-function(Category1="Apple Braeburn", Category2="Apple Crimson S
now",img_path_dir=img_path_dir, rand_err=20, details=TRUE){
#browser()
Train_cate1<-ExtractFeature(FolderName=Category1, details=details, img_path_dir = img
_path_dir)
Train_cate2<-ExtractFeature(FolderName=Category2, details=details, img_path_dir= img_
path_dir)
data=rbind(Train_cate1, Train_cate2)

Y_var<-ifelse(data$Y==Category1, 1, 0)

vv<-round(runif(rand_err, 1,length(Y_var) ))
Y_var[vv]= 1- Y_var[vv]
data$Y= Y_var
return(data)
}
```

Function to Extract the fearture from testing set.

```
ExtractFeature_test<-function(FolderNameLoc){
  #img_path_dir="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Training/"
  #img_path1="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Training/Orang
e/0_100.jpg"

  #img_path_dir
  img_path<-paste0(FolderNameLoc)
  #print(img_path)
  img <- tryCatch(expr = {readImage(img_path)}),
    error = function(e){
      # (Optional)
      # Do this if an error is caught...
      return(NULL)
    }
  )

  #browser()
  if(!is.null(img)){
    val_color_feature<- (c(Extract_col_feature(img)))
    val_color_feature=rbind(val_color_feature,val_color_feature)
  }

  #Y=rep(x = FolderName, dim(val_color_feature)[1])
  #val_color_feature
  return(data.frame(val_color_feature)[1,])
}
```

Sample Images of the objects that we are considering (Just for Checking if the

Folders are set correctly)

```
CategoryLabels=data.frame(Category=c("Apple Braeburn","Apple Crimson Snow" ), Respons  
eInd=c(1,0)) # Means Apple Braeburn===1, Apple Crimson Snow===0  
img_path=paste0(dir_location, "Training/",CategoryLabels$Category[1],"/1_100.jpg")
```

```
img<-readImage(img_path)  
imageShow(img)
```



```
img_path=paste0(dir_location,"Training/",CategoryLabels$Category[2],"/1_100.jpg")  
img<-readImage(img_path)  
imageShow(img)
```



Creating the Training Dataset by extracting the features by reading all the samples/images from the Data folder

```
#CategoryLabels=data.frame(Category=c("Apple Braeburn","Apple Crimson Snow" ), ResponseInd=c(1,0)) # Means Apple Braeburn===1, Apple Crimson Snow===0
#img_path=paste0(dir_location, "Training/",CategoryLabels$Category[1],"/1_100.jpg")
Training_dir_location<-paste0(dir_location, "Training/")
```

```
data<-Create_Training_Data(Category1 = CategoryLabels$Category[1], Category2 =CategoryLabels$Category[2],img_path_dir =Training_dir_location, details = FALSE)
dim(data)
```

```
## [1] 936 13
```

```
#data<-Create_Train_Data(rand_err = 15, details = FALSE)
#dim(data)
```

Fit a logistic regression with Y=Response

```
fit<-glm(Y~., data=data, family="binomial")  
#cbind(fit$fitted.values, data$Y)
```

**Prediction on a data from the testing set:
Predict the category of the following object
we want to check if a specific image is Apple
Crimson Snow or Apple Braeburn?**

```
Test_Folder_Loc= paste0(dir_location, "Test/Apple Crimson Snow/80_100.jpg")  
  
#FolderNameLoc="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Test/Apple B  
raeburn/80_100.jpg"  
img<-readImage(Test_Folder_Loc);imageShow(img)
```



```
#FolderNameLoc="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Test/Apple Golden 1/63_100.jpg"
```

```
newData_feature=ExtractFeature_test(Test_Folder_Loc)
predictedProb<-predict.glm(fit, newdata=newData_feature, type="response")
paste0("The predicted probability that the selected image is:",CategoryLabels$Category[1],"( comparing between ", paste0(CategoryLabels$Category, collapse=" and "), ") is ", predictedProb )
```

```
## [1] "The predicted probability that the selected image is:Apple Braeburn( comparing between Apple Braeburn and Apple Crimson Snow) is 0.0487164534760055"
```

```
predictedProb_1=data.frame(predictedProb, 1-predictedProb)
colnames(predictedProb_1)=CategoryLabels$Category
rownames(predictedProb_1)="Predicted Group Probability"
predictedProb_1
```

```
##                                Apple Braeburn Apple Crimson Snow
## Predicted Group Probability    0.04871645          0.9512835
```

APPLE AND ORNAGE Dataset

Sample Images of the objects that we are considering

```
CategoryLabels=data.frame(Category=c("Orange","Apple Braeburn" ), ResponseInd=c(1,0))
#Orange===1, and Apple Braeburn===0
img_path=paste0(dir_location, "/Training/",CategoryLabels$Category[1],"/1_100.jpg")
img<-readImage(img_path)
imageShow(img)
```



```
img_path=paste0(dir_location, "/Training/",CategoryLabels$Category[2],"/1_100.jpg")  
img<-readImage(img_path)  
imageShow(img)
```




Creating the Training Dataset by extracting the features by reading all the samples/images from the Data folder

```
CategoryLabels=data.frame(Category=c("Orange","Apple Braeburn" ), ResponseInd=c(1,
0))
#Orange===1, and Apple Braeburn===0
Training_Folder_Loc=paste0(dir_location, "Training/")
data_o<-Create_Training_Data(Category1 = CategoryLabels$Category[1], Category2 =CategoryLabels$Category[2], img_path_dir = Training_Folder_Loc, details = FALSE)
dim(data)
```

```
## [1] 936 13
```

```
fit_o<-glm(Y~., data=data_o, family="binomial")
#Estimated_prob<-cbind(fit$fitted.values, data$Y)
```

Predict the category of the following object

```
FolderNameLoc=paste0(dir_location, "Test/Apple Braeburn/80_100.jpg")
img<-readImage(FolderNameLoc);imageShow(img)
```



```
#FolderNameLoc="/Users/subhadippal/Downloads/Fruit-Images-Dataset-master/Test/Orange/6_100.jpg"
#img<-readImage(FolderNameLoc);imageShow(img)

newData_feature=ExtractFeature_test(FolderNameLoc)
predictedProb<-predict.glm(fit_o, newdata=newData_feature, type="response")
paste0("The predicted probability that the selected image is:",CategoryLabels$Category[1],"( comparing between ", paste0(CategoryLabels$Category, collapse=" and "), ") is ", predictedProb )
```

```
## [1] "The predicted probability that the selected image is:Orange( comparing between n Orange and Apple Braeburn) is 0.0059485586682921"
```

```
predictedProb_1=data.frame(predictedProb, 1-predictedProb)
colnames(predictedProb_1)=CategoryLabels$Category
rownames(predictedProb_1)="Predicted Group Probability"
predictedProb_1
```

```
##                                Orange Apple Braeburn
## Predicted Group Probability 0.005948559          0.9940514
```